Teaching MLB Players How to Use ABS To Their Advantage

Grant Suchecki

Introduction

The introduction of the Automated Balls and Strikes (ABS) challenge system in the MLB has created a large learning curve for teams entering the 2026 season. Teams are not allowed to signal to players when they should challenge or not during a game. The Milwaukee Brewers were warned to stop aiding players in challenge decisions. It is critical that players understand the importance of challenges and the proper context in which they should be used. Teams are only allowed two challenges per game. If they lose a challenge, they do not get it back. Meaning that they should be used sparingly, in situations where players know that it would be advantageous to do so. Players should avoid burning crucial challenges at the beginning of games. Instead of predicting whether a challenge will be overturned, players should be taught when the game context is correct to use a challenge.

Related Work

ABS is a brand new system in the MLB outside of its previous year's testing in the minor leagues. Most of the literature created has been from the MLB itself on stattracking and leaderboards for players and teams. Much of the work on teaching challenge strategy to players is internal to each baseball club. An extensive search showed that the only scholarly article to be written on the use of ABS was written about the Korean Baseball Organization (KBO League), which automated ball and strike calls, rather than auditing umpires' real time calls. The work does touch on player adaptation to the system, but is not necessarily helpful in determining challenge behavior, since every call is automated in the KBO (Lee, K. et al., 2025).

Each team's use of teaching mechanics for the system is unique. However, I sought out to provide a broad teaching platform that all teams can use in the league. Instead of singling out different players for their challenging habits, the goal was to find the proper context in which players should be using these challenges.

Data Description

The data has been collected over the start of the MLB season from the Sabermetrics R package (Powers, 2024). The data runs up until April 20th, attempting to get as many pitches into the dataset as possible. There have been over 100,000 pitches in the MLB season so far, with nearly 1,000 of them being challenged pitches. The dataset includes everything from pitch location, to pitch event, pitch type, and several game context factors that all go into determining if a player challenges a pitch. Flags that include if a pitch was challenged or overturned are also included, with the challenge flag being the main dependent variable that the model is based around.

This data was handled mainly in R. Some early visualizations were created using downloaded CSVs from the MLB Statcast leaderboard and BaseballSavant pages on the MLB website. These datasets did not include pitch by pitch data. The current dataset has different pitch location data for future use. Rather than a specific strike zone coordinate, the data from Sabermetrics R includes release location, velocity, and rotations, leaving the user to calculate the final position of the ball.

Methods

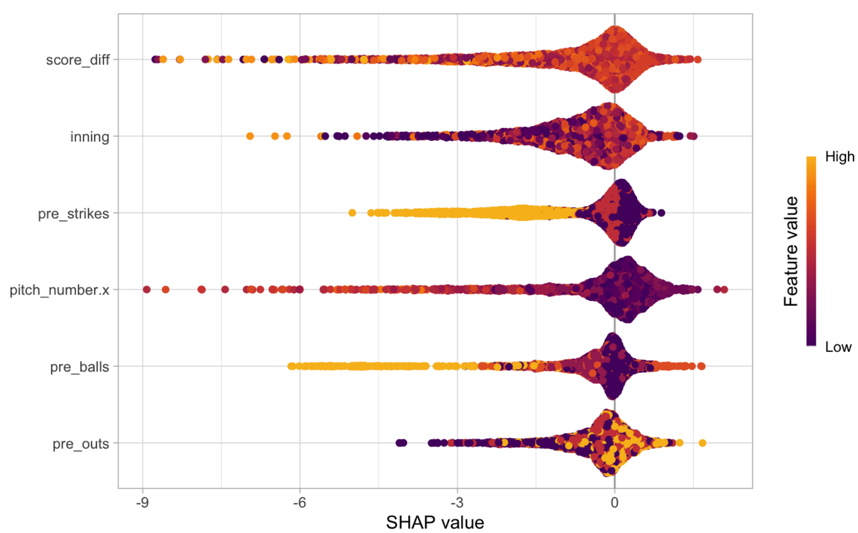
The primary dependent variable of this model was abs_challenged, a binary marker that demonstrated if a pitch was included in a challenge by a player. The main goal was not to determine if a call was being overturned, but describe what game context warrants a challenge attempt. An engineered variable was also created called score_diff, aka score differential. This is based on the score of the home team. This was created to help determine if the difference in scores has a tell in when teams challenge.

The first iteration of the model used a logistic regression model, including pre-pitch data, such as balls, strikes, outs, innings pitched, score differential, as well as the current inning. This inadvertently was used to measure abs_challenges over a game's length. Further, an XGBoost model was run as the primary model for the problem. Each of the same variables from the logistic regression were used on the XGBoost. While there aren't many variables present in the data, XGBoost is built to handle larger, more complex datasets for machine learning evaluation. I also used XGBoost in order to get Shapley graphs mentioned in the results section below. A ratio was added to the scale weights section of the model that aimed to balance the drastic gap of non-challenged pitches to challenged pitches, which previously sat at about 100:1.

Results

The logistic regression model, as mentioned above, helped describe the linear pattern of challenges during a game. Inning was shown to be the most significant of all of the variables included in the regression. This seemed to align with the previous belief that was held before running the model, that smarter teams either hold or challenge smartly in order to maintain challenges by the final innings of the game. The linear regression also placed significance on pre-pitch balls, outs, and the pitch number of the pitcher on the mound. This was an interesting insight as it showed that defensive players are most likely challenging in order to preserve a pitcher's pitch count during a game, especially if they want to avoid extending an at-bat or an inning.

The XGBoost model provided important insights that leaned more on the full game context first rather than the pitch context. The top two features marked by the model were score differential and inning. The remaining variables in rank of importance are shown in the SHAP visualization



on the left. These followed expectations, as tighter games are going to see more challenges than blowouts. Late inning challenges are still shown to be favorable by teams as well. The model was able to generate an accuracy of almost 73%. The Area Under The Curve-PR came out to .9007, predicting that a random pitch would be challenged 90% of the time. This helps describe random challenges as well, the 10% that were challenged in unusual game contexts.

Actions

MLB teams can implement many different strategies using this model. Not only can it describe how challenges are currently occurring, but it can show lessons for how to use the system advantageously.

I would stress the importance of teams waiting to use their challenges until late in the game, or if they cannot wait, only using them on pitches where the call made by the umpire was almost egregiously incorrect. By doing this, teams can put themselves in a better position to be able to make challenges when game context is of utmost importance. Later games mean more high pressure calls, meaning less room for error. Umpires and players are both under pressure in these situations.

Another challenge lesson I would teach is using challenges when games are closer in the later innings. This follows the previous lesson, but includes that score differential is an important context according to the XGBoost's feature importance list. Teams can also use their challenges in near blowouts, for example, if a losing team is attempting to make a comeback during a game, they could afford to burn a challenge to gain momentum.

One of the most important lessons that players can learn from this is the importance of flipping strikes into walks. Focusing on getting on base is shown by both models to be one of the more important reasons for challenging. Players would rather flip a walk than a strikeout, unless the flipped strikeout is turned into a walk. Extending an at-bat just to strike out the following pitch is unproductive; flipping a 3-2 count into a walk is.

Future Work & Conclusion

For future work on this project, I would like to shift the model to work with pitch location data as well as the possibility of different pitch types seeing more challenges. For example, breaking balls that catch the corner may be challenged more than a fastball that seems to stay in the same spot the entire way through the strike zone. It also shows what pitches umpires are more accurate on. Pitch location data would be incredibly context as there would be a need for perceived distance from the strike zone. Raw location data would be unhelpful for showing why a batter would challenge since they cannot necessarily perceive the zone the same way that the cameras that power ABS can.

This model allows for teams to take a great look into what game context drives challenges in the new era of ABS in the MLB. Lessons can be learned from the model, such as the importance of late-game strategy as well as pure at-bat strategy. Saving challenges can save teams games in the long run. While losing one game out of 162 doesn't seem like very important, those games add up and one challenge can decide if a team will clinch a playoff spot at the end of the season.

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